Windowed Hankel KEM: A Compact Lattice-Based KEM from Truncated Linear Convolution

Marcos Mendoza III
Independent Researcher

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Abstract

We present *Windowed Hankel KEM*, a key encapsulation mechanism based on a structured variant of the Learning with Errors (LWE) problem. The public matrix is a Hankel matrix generated by a single pseudorandom sequence. Equivalently, the public key is a noisy *middle product*: the fully supported coefficients of a linear (non-cyclic) convolution of the public sequence with a short secret.

This structure lies between unstructured LWE, as used by FrodoKEM, and ring- or module-LWE, as used by ML-KEM. It compresses the public key without reducing products modulo a fixed polynomial. Middle-Product LWE (MP-LWE), introduced by Roşca, Sakzad, Steinfeld and Stehlé, admits reductions from Polynomial-LWE over a large family of rings. However, our instance does not satisfy the hypotheses of those reductions: it uses a single long public sequence, a short binomial secret, and a narrow error distribution. We therefore claim no reduction, and we assess security by concrete lattice-attack estimates.

With $n = 640$, a window of $W = 128$ coordinates, $q = 2^{12}$, and binomial noise ($\eta = 2$), the scheme encapsulates a 256-bit message by 4-PAM modulation. Its public key is 1,196 bytes and its ciphertext 1,356 bytes, both including 12-byte format headers. A Chernoff bound gives a decryption failure rate below $2^{-134.1}$. Treating the instance as unstructured LWE, the cheapest attack found by the lattice estimator is the dual hybrid, with BKZ block size $\beta = 468$ and an estimated cost of $2^{137.1}$ in the Core-SVP model. Under the same model and tool, Kyber-512 costs $2^{115.5}$. A Python implementation with ARMv8 NEON kernels completes a handshake in 0.34 ms on an Apple M3 Max.

Keywords: post-quantum cryptography, key encapsulation, learning with errors, middle-product LWE, Hankel matrices.

1 Introduction

Lattice-based key encapsulation mechanisms (KEMs) occupy a spectrum of algebraic structure.

At one end, unstructured LWE [12] uses a uniformly random public matrix $A \in \mathbb{Z}_q^{m \times n}$ and is backed by worst-case reductions from standard lattice problems. FrodoKEM [4] follows this approach. Its Category 1 parameter set transmits 19,336 bytes per exchange (public key plus ciphertext).

At the other end, ring- and module-LWE schemes such as ML-KEM [5, 10] replace A by matrices of polynomials in $\mathbb{Z}_q[X]/(X^n + 1)$. A single ring element then describes an $n \times n$ block. This reduces ML-KEM-512 to 1,568 bytes per exchange and enables NTT-based multiplication. The resulting algebraic structure has been studied extensively; see [11] for a survey.

This paper studies an intermediate structure. Here the public matrix is a *Hankel* matrix $H_{i,j} = a_{i+j}$ generated by a single sequence a. Multiplying by H computes a contiguous slice of the coefficients of the polynomial product $a(X)\, s(X)$ in $\mathbb{Z}_q[X]$, without reduction modulo any polynomial.

The choice of slice matters. The outermost coefficients of a full linear convolution depend on only a few secret coordinates; for instance, $c_0 = a_0 s_0 + e_0$. Publishing them would let an attacker recover a small secret coordinate by coordinate. We therefore publish only the *fully supported* coefficients, each of which depends on all n secret coordinates. This is precisely the *middle product* of Roşca, Sakzad, Steinfeld and Stehlé [13], who introduced Middle-Product LWE (MP-LWE). Steinfeld, Sakzad and Zhao later designed the Titanium cryptosystem around it [14, 15].

The MP-LWE reductions cover a specific sample distribution: many samples, each with a fresh short public polynomial; a long uniform secret; and Gaussian noise of sufficient width. Titanium uses parameters compatible with these reductions; its 128-bit KEM uses a 16,352-byte public key and a 3,552-byte ciphertext [14]. Our design deliberately departs from that distribution in order to be compact. It uses a single long public sequence, a short binomial secret, and narrow binomial noise. Its security is therefore a heuristic assumption, supported by concrete cryptanalysis rather than by a reduction.

Contributions.

1. A KEM based on windowed Hankel (middle-product) products, with a 128-coordinate window that carries a 256-bit message by 4-PAM modulation, and a modulus $q = 2^{12}$ that reduces all arithmetic to masking (Section 3).

2. A precise account of the scheme's relation to MP-LWE, including why the known reductions do not apply to the chosen parameters (Section 2.2).

3. A decryption-failure analysis with an exact Chernoff bound, and concrete estimates for primal, dual and hybrid lattice attacks (Section 5).

4. An IND-CCA KEM via the Fujisaki–Okamoto transform with implicit rejection [6], and an ARMv8 NEON implementation (Section 6).

2 Preliminaries

2.1 Hankel operators and the middle product

Throughout, $q = 4096$ and $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$.

Definition 1 (Hankel matrix)**.** A matrix $H \in \mathbb{Z}_q^{m \times n}$ is a *Hankel matrix* if $H_{i,j} = a_{i+j}$ for some sequence $a = (a_0, \ldots, a_{m+n-2}) \in \mathbb{Z}_q^{m+n-1}$.

Write $a(X) = \sum_t a_t X^t$ and $s(X) = \sum_{j<n} s_j X^j$. Then $(Hs)_i = \sum_j a_{i+j} s_j$. This is the coefficient of X^{i+n-1} in the product $a(X)\,\tilde{s}(X)$, where $\tilde{s}$ is the reversal of s. Hence Hs is a contiguous slice of a polynomial product in $\mathbb{Z}_q[X]$.

Choosing a of length $m + n - 1$ ensures that every published coordinate depends on all n entries of s. This is the *middle product* of [13]. Hankel matrices are not closed under multiplication and do not form a ring. Nevertheless, the scheme is polynomial-structured: it uses polynomial multiplication in $\mathbb{Z}_q[X]$, but without reduction modulo a fixed polynomial.

2.2 Relation to Middle-Product LWE

In MP-LWE [13], each sample consists of a fresh uniformly random short polynomial a_i together with the middle product of a_i and a fixed secret s of roughly twice the length, plus Gaussian noise. Roşca et al. reduce Polynomial-LWE, over any polynomial in a large family, to MP-LWE. Consequently, MP-LWE is at least as hard as Polynomial-LWE for the hardest member of that family. Steinfeld et al. tighten this reduction and use it to set Titanium's parameters [15].

The distribution used by Windowed Hankel KEM differs in three respects.

1. A *single* long public sequence a of length $2n + W - 1$ is used, rather than many fresh short samples.
2. The secret is *short*, drawn from CBD_2, rather than long and uniform.
3. The error has standard deviation 1, far below the width the reductions require.

The reductions of [13, 15] therefore do not apply, and we do not claim that the scheme inherits their guarantees. We regard the hardness of recovering s from $(a, Hs + e)$, with H windowed Hankel, as a heuristic assumption. Section 5 evaluates it with concrete lattice attacks.

2.3 Power-of-two modulus

With $q = 2^{12}$, reduction modulo q is a bitwise mask. Uniform elements are obtained by masking 16-bit words of SHAKE256 [9] output, without rejection. Two 12-bit coefficients pack exactly into three bytes, so every byte string of the correct length decodes to canonical coefficients. The cost is that NTT-based multiplication is unavailable. Products are computed directly in $O(mn)$ operations (Section 6).

2.4 4-PAM modulation

A 256-bit message is carried by $W = 128$ coordinates, two bits per coordinate. Symbol $\mu \in \{0, 1, 2, 3\}$ maps to $\mu \cdot 1024 \in \mathbb{Z}_{4096}$.

Proposition 1 (Decoder)**.** Let $w = 1024\mu + e \bmod 4096$ with $-512 \le e \le 511$. Then $\hat{\mu} = ((w + 512) \bmod 4096) \gg 10$ equals μ.

Proof. We have $w + 512 \equiv 1024\mu + (e + 512) \pmod{4096}$ with $0 \le e + 512 \le 1023$. Hence $\lfloor ((w + 512) \bmod 4096)/1024 \rfloor = \mu$, and division by 1024 is a right shift by 10. □

3 The Scheme

The parameters are $n = 640$, $W = 128$, $m = n + W = 768$, $q = 4096$ and $\eta = 2$. The public sequence has length $L_a = 2n + W - 1 = 1407$. $H \in \mathbb{Z}_q^{m \times n}$ denotes the Hankel matrix of a. CBD_2 samples values in $\{-2, \ldots, 2\}$ with probabilities $(1, 4, 6, 4, 1)/16$, so its variance is 1.

Algorithm 1 CPA key generation

Require: seed $\in \{0, 1\}^{512}$

$seed_a \leftarrow seed[0{:}32]$; $seed_e \leftarrow seed[32{:}64]$
$a \leftarrow$ (SHAKE256$(seed_a, 2L_a)$ read as 16-bit words) mod 4096
$\beta \leftarrow$ SHAKE256$(seed_e \,\|\, \texttt{"_hankel_noise"}, n + m)$
$s \leftarrow \mathrm{CBD}_2(\beta[0{:}n])$; $e \leftarrow \mathrm{CBD}_2(\beta[n{:}n + m])$
$b \leftarrow Hs + e \bmod q$
return $pk = (seed_a, b)$, $sk = s$

Algorithm 2 CPA encryption of a 32-byte message M with coins c

Require: $pk = (seed_a, b)$, message M, coins $c \in \{0, 1\}^{256}$

Regenerate a from $seed_a$
$\gamma \leftarrow$ SHAKE256$(c \,\|\, \texttt{"_hankel_encaps_coins"}, n + m + W)$
$r \leftarrow \mathrm{CBD}_2(\gamma[0{:}n])$; $e_1 \leftarrow \mathrm{CBD}_2(\gamma[n{:}n + m])$; $e_2 \leftarrow \mathrm{CBD}_2(\gamma[n + m{:}n + m + W])$
$u \leftarrow Hr + e_1 \bmod q$
$K[w] \leftarrow \sum_{j=0}^{n-1} r_j \, b_{w+j} \bmod q$ for $0 \le w < W$
$v \leftarrow K + \mathrm{PAM}(M) + e_2 \bmod q$
return (u, v)

Algorithm 3 CPA decryption

Require: $sk = s$, ciphertext (u, v)

$K'[w] \leftarrow \sum_{k=0}^{n-1} s_k\, u_{w+k} \bmod q$ for $0 \le w < W$

return Decode$(v - K' \bmod q)$

Correctness. Expanding b and u gives

$$K[w] = \sum_{j,k} a_{w+j+k}\, r_j s_k + \sum_j r_j e_{w+j}, \qquad K'[w] = \sum_{j,k} a_{w+j+k}\, r_j s_k + \sum_k s_k e_{1,w+k}.$$

The double sums coincide because each depends only on $j + k$. Therefore

$$v - K' = \mathrm{PAM}(M) + \underbrace{\sum_{j=0}^{n-1} r_j e_{w+j} - \sum_{k=0}^{n-1} s_k e_{1,w+k} + e_{2,w}}_{e_{\mathrm{dec}}[w]} \pmod q,$$

and decryption succeeds whenever $|e_{\mathrm{dec}}[w]| < 512$ for all w.

The structure $K \approx r^\top H s$ is the standard one for LWE-based key exchange [7]. The term e_2 is essential. Without it, the W coordinates of v would give an attacker W noiseless linear equations in the ephemeral secret r.

IND-CCA KEM. We apply the Fujisaki–Okamoto transform with implicit rejection, $\mathrm{QFO}^{\not\perp}_m$ [6].

- *Encapsulation* derives $(\bar{K}, c) \leftarrow \mathrm{SHAKE256}(M \,\|\, \mathrm{SHA256}(pk), 64)$ for uniform M, and encrypts M with coins c.
- *Decapsulation* decrypts to M' and re-encrypts. It compares the result with the received ciphertext in constant time, and selects either $\bar{K}'$ or a secret rejection value without branching.
- The shared secret is $\mathrm{SHAKE256}(K_{\mathrm{sel}} \,\|\, \mathrm{SHA256}(ct), 32)$.

Malformed ciphertexts are parsed without data-dependent exceptions: length, header and parameter checks are combined with bitwise operations.

4 Parameters and Sizes

Parameter	Value
Dimension n	640
Window W / published coordinates $m = n + W$	128 / 768
Modulus q	$4096 = 2^{12}$
Secret and error distribution	CBD_2 (variance 1)
Public sequence length $2n + W - 1$	1,407
Message	256 bits (4-PAM over W coordinates)
Public key	1,196 B
Ciphertext	1,356 B
Secret key	1,868 B

Table 1: Parameters of Windowed Hankel KEM. The public key consists of a 12-byte header, a 32-byte seed and 768 coefficients of 12 bits; the ciphertext of a 12-byte header and $768 + 128$ coefficients of 12 bits.

Table 2 compares wire sizes with LWE-based KEMs at the 128-bit level. For a like-for-like comparison, the Windowed Hankel figures in that table exclude our 12-byte format headers.

Scheme	Structure	Public key	Ciphertext	Total
ML-KEM-512 [10]	module-LWE	800 B	768 B	1,568 B
Windowed Hankel KEM (this work)	middle product	1,184 B	1,344 B	2,528 B
Titanium-CCA-Std128 [14]	MP-LWE	16,352 B	3,552 B	19,904 B
FrodoKEM-640 [4]	unstructured LWE	9,616 B	9,720 B	19,336 B

Table 2: Wire sizes of LWE-based KEMs targeting the 128-bit security level.

The ciphertext is transmitted uncompressed. Rounding u and v to fewer bits, as ML-KEM does, would shrink it further but requires a revised failure analysis. We leave this to future work.

5 Security Analysis

5.1 Lattice attacks

Following the Core-SVP methodology [2, 3], we first treat the public key as an instance of unstructured LWE with $n = 640$, up to $m = 768$ samples, $q = 4096$, and secret and error of standard deviation 1. The same estimate applies to the ciphertext component u, whose secret is r.

Estimates. We use the lattice estimator [1] (commit `53da598`, August 2026) under two cost models for BKZ: the Core-SVP model of [3], which charges $2^{0.292\beta}$ classically for one call to an SVP oracle in dimension β, and the more detailed model of MATZOV [8]. Table 3 lists the results. For calibration, we ran the same version of the estimator on its built-in Kyber-512 parameters; the Kyber-512 figures it produces coincide with those in the estimator's documentation.

	Windowed Hankel KEM				Kyber-512	
	Core-SVP		MATZOV		Core-SVP	MATZOV
Attack	β	cost	β	cost	cost	cost
Primal (uSVP)	488	$2^{142.5}$	488	$2^{166.8}$	$2^{118.6}$	$2^{143.8}$
Primal (BDD)	488	$2^{143.2}$	471	$2^{163.1}$	$2^{119.4}$	$2^{140.2}$
Dual	507	$2^{148.0}$	509	$2^{173.7}$	$2^{123.8}$	$2^{149.9}$
Dual hybrid	468	$2^{137.1}$	459	$2^{160.3}$	$2^{115.5}$	$2^{139.7}$
Coded-BKW	—	—	—	$2^{207.3}$	—	$2^{178.8}$

Table 3: Lattice-estimator results for the public key ($n = 640$, $m = 768$, $q = 4096$, CBD_2 secret and error) and for Kyber-512, treated as unstructured LWE. Costs are the estimator's total cost (`rop`), including guessing in the hybrid attack. Coded-BKW is costed without lattice reduction, so it appears only once per scheme.

The dual hybrid attack is the cheapest in both models. In the Core-SVP model it uses $\beta = 468$, so its lattice-reduction cost alone is $2^{0.292 \cdot 468} = 2^{136.7}$ classically and $2^{0.265 \cdot 468} = 2^{124.0}$ quantumly. The primal attack uses $\beta = 488$ in a lattice of dimension 1,194, at a classical Core-SVP cost of $2^{142.5}$ and a quantum one of $2^{129.3}$. Kyber-512 reports a classical Core-SVP cost of 2^{118} for its Category 1 claim [5], in agreement with the estimator's primal figure. In every row of Table 3, the Windowed Hankel instance costs at least 20 bits more than Kyber-512 under the same model; the smallest margin, 20.6 bits, is for the dual hybrid attack in the MATZOV model. These figures assume that the Hankel structure gives the attacker no advantage.

Effect of the Hankel structure. These estimates ignore the structure of H, and an attacker might exploit it. Its rows are shifts of one another, and the same sequence a generates both b and u, much as ML-KEM reuses its matrix A. We are not aware of an attack that improves on the unstructured estimates for this single-sample Hankel setting. However, the setting has received little study. Steinfeld et al. analyze lattice attacks on MP-LWE taking its Toeplitz structure into account [15]. Adapting that analysis to our parameters is the most important open question for this scheme.

5.2 Decryption failure rate

Theorem 1. The decryption failure rate satisfies $\mathrm{DFR} \le 2^{-134.1}$.

Proof. For each window coordinate w, the error $e_{\mathrm{dec}}[w]$ is a sum of 1,281 independent terms. Of these, 1,280 are products $X \cdot Y$ of independent CBD_2 variables, and one is a CBD_2 variable ($e_{2,w}$). Each term has mean 0 and variance 1. Hence $\sigma = \sqrt{1281} \approx 35.79$, and the threshold 512 lies 14.31σ from the mean.

Evaluating the exact moment-generating functions of these distributions, the Chernoff bound gives $\Pr[e_{\mathrm{dec}}[w] \ge 512] \le 2^{-142.1}$. By symmetry, $\Pr[|e_{\mathrm{dec}}[w]| \ge 512] \le 2^{-141.1}$. A union bound over the $W = 2^7$ coordinates gives $\mathrm{DFR} \le 2^{-134.1}$. □

In 500 end-to-end IND-CCA exchanges (64,000 decoded coordinates), the largest observed $|e_{\mathrm{dec}}|$ was 131. The mean of $|e_{\mathrm{dec}}|$ was 28.4 and its standard deviation 21.4, close to the values of 28.5 and 21.6 predicted by a normal approximation. These observations are consistent with the bound; an experiment of this size cannot itself confirm a failure rate of 2^{-134}.

6 Implementation

We provide a Python reference implementation and an ARMv8 NEON implementation of the Hankel products and of decoding.

Accumulation without intermediate reductions. The forward product $\sum_{j<n} a_{i+j} s_j$ multiplies 16-bit sequence elements by 16-bit signed secret coefficients and accumulates the products in 32-bit lanes using `vmlal_s16`. The largest possible magnitude is $640 \cdot 4095 \cdot 2 = 5{,}241{,}600$, far below 2^{31}. Each output therefore needs only a single reduction (a mask with `0x0FFF`).

Branch-free decoding and selection. The 4-PAM decoder is applied to eight coordinates at a time:

```
uint16x8_t vw = vld1q_u16(in_w + i);
vw = vaddq_u16(vw, vdupq_n_u16(512));
vw = vandq_u16(vw, vdupq_n_u16(0x0FFF));
vw = vshrq_n_u16(vw, 10);
```

The ciphertext comparison and the implicit-rejection selection use bitwise select operations.

Performance and test vectors. The KEM is orchestrated in Python, with hashing, sampling and serialization in Python and NumPy; the NEON code computes the Hankel products and decoding. On an Apple M3 Max (Python 3.13, NumPy 2.3), the median latencies over 2,000 fresh key pairs and ciphertexts are 0.07 ms for key generation, 0.11 ms for encapsulation and 0.15 ms for decapsulation, and 0.34 ms for a full handshake. These figures include Python overhead and are not representative of a fully native implementation. We do not report timings for other schemes, because they were not measured on the same platform under the same conditions. Fifteen deterministic known-answer test vectors, generated with a custom SHAKE-based generator rather than the NIST DRBG, are reproduced exactly by the implementation.

7 Limitations

- *No reduction.* The instance lies outside the hypotheses of the MP-LWE reductions (Section 2.2). Its security rests on concrete estimates for unstructured LWE, together with the assumption that the Hankel structure does not help an attacker.
- *Unstructured estimates.* All attack costs in Section 5 treat the instance as unstructured LWE. Attacks that exploit the Hankel structure have not been analyzed.
- *No native implementation.* Only the arithmetic kernels are native; the reported timings include Python overhead.
- *Quadratic arithmetic.* Without an NTT, multiplication costs $O(mn)$ operations, about 4.9×10^5 multiply-accumulates at $n = 640$. This is acceptable here, but it scales less favorably than ring-based schemes at higher security levels.
- *Cryptanalytic maturity.* The single-sample Hankel setting has received little study. Independent cryptanalysis is needed before any use beyond research.

8 Conclusion

Windowed Hankel KEM publishes noisy middle products of a single pseudorandom sequence with a short secret. The result is a 2.5 KB exchange with a decryption failure rate below 2^{-134}, and estimated lattice-attack costs at least 20 bits above those of Kyber-512 under the same cost models. The design gives up the MP-LWE reductions in exchange for compactness. Whether the Hankel structure can be exploited in this regime is the central open question.

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